PA consistency machines: current comparison and further reduction experiments

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Abstract

This investigation compares the current PA constructions and every incoming archive at a fixed ProveIt revision. It does *not* construct a verified binary Turing machine with fewer than 196 working states. The requested record reduction remains unresolved.

The comparison covers 23 top-level archives, 36 distinct archive payloads after following nested historical inputs, and 1,223 literal transition-table members. The designated 196-state corpus contains fifteen different rooted transition graphs; a prior fourteen-table census omits a later twelve-rotation layout. Five 195-state files represent four rejected candidate constructions. We supply an updated portable census, exact tables and provenance, a checked family of equivalent separation gates, and new bounded physical and grammar searches. A restricted formula-size calculation establishes four shortest implication-only separation matrices. This is a local mathematical result, not a state-count lower bound.

The proof boundary is explicit throughout. The imported formal arithmetic endpoint is $\mathrm{PA} \vdash \mathrm{Nonhalt}(M_{196}) \to \mathrm{Con}(\mathrm{PA})$; the requested internal converse is a separate obligation. An external nonhalting theorem, an instruction simulator, and a proof in PA are different deliverables.

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1 The target and the outcome

The requested target is an explicit machine M satisfying

$$\#\text{WorkingStates}(M) \leq 195, \qquad \text{PA} \vdash \big(\text{Nonhalt}(M) \leftrightarrow \text{Con}(\text{PA})\big).$$

We preserve the conventions of the compared constructions: one two-way infinite tape, alphabet $\{0, 1\}$, blank input, one-cell left or right moves, and a distinguished halt control excluded from the working-state count. Changing the alphabet or supplying a nonblank program tape would change the comparison problem.

Neither part of a new sub-196 construction is established here. The archive includes the existing transition tables as references and the failed 195-state tables as clearly marked diagnostics. None is presented as a successful answer to the target. The value of the packet is a more complete comparison, executable checks, a new locally proved gate family, and explicit records of the additional search space explored.

The source snapshot is ProveIt commit

`3268f238cb6be4b9c68b6cddaf289f173f77566e`.

The imported James Chang trees are PA196 revision `407cdc356f6b586635e71ac92be394255f4bfea4` and PA209 revision `69d98241a721ead20b4f0b17df9062950b40fbfe`. All source claims below refer to these fixed inputs, not an indefinitely moving branch.

2 What was compared

2.1 The main families

Family	States	Distinguishing construction and status
James Chang PA209	209	Direct predecessor of PA196; 3,948 primitive source sites.
Original U208	208	A distinct U209 lineage, 3,909 sites, with a history-sensitive local fold. Nine original presentations form two rooted graph classes.
U208 helper variants	208	Eighteen additional helper choices give eighteen additional rooted graph classes.
Published PA196	196	3,812 sites; transposed quantifier coding, six axiom heads, guarded equality vacuity, and shared readers.
Other designated PA196 layouts	196	Fourteen more distinct tables, including two experimental MP variants. Source changes and proof scopes differ.
Negation / combined MP	197	Shorter source programs of 3,596 and 3,573 sites, respectively; no state record.

U208's selected U209 parent is not James Chang's PA209 merely because both have 209 states. Their identities must be compared through their source and transition data. Likewise, a repeated ZIP or a dense renumbering does not constitute an independent construction.

The current PA196 paper decomposes its 196 states as 36 engine states, 13 leaf-routing states and 147 reader states representing 159 distinct source-tree nodes [1]. Its six complementary reader-pair folds are already included. They cannot be subtracted a second time.

2.2 Complete archive and table inventory

The archive scanner parses data without executing programs found in the archives. It follows nested ZIPs and deduplicates archive payloads by their SHA-256 digest. The resulting counts are:

Quantity	Count
Top-level incoming archives	23
Distinct archive payloads, including nested inputs	36
Maximum nesting depth	7
Literal `.tm` members in those payloads	1,223
Byte-distinct table files	489
Distinct parsed transition tables	484
Unparsable `.tm` members	0
Designated 196-state tables / rooted graph classes	15 / 15
Designated U208 presentations / rooted graph classes	27 / 20
195-state diagnostic files / candidate constructions	5 / 4

The broad table counts include duplicates, ancestors, experimental files, rejected candidates and two small synthetic controls. They are not counts of accepted consistency machines. The nested historical lineage adds only 209-state or larger tables; it supplies no overlooked smaller construction. The inventory and the per-archive comparison matrix are retained under `census/`. Appendix A lists all 23 top-level inputs.

For the fifteen designated tables, the new checker verifies the exact file digests, both binary rows of every working state, graph reachability from the designated start, and pairwise nonisomorphism preserving the two read symbols and each write/move action. It canonicalizes reachable states by deterministic exploration from the start. Every one has 392 rows and 196 graph-reachable working states. Graph reachability permits arbitrary sequences of read symbols; it does not assert that every state occurs on the single blank-tape execution.

2.3 The fifteen 196-state graphs

Representative	Scope of the supplied justification
Published PA196	Imported formal development; the precise arithmetic endpoint is discussed in Section 6.
A/D register swap	Reported uniform symbolic contracts and startup.
MP result in H	Experimental command change; bounded frame checks and separate compiler obligations.
MP without zero push	Experimental command change; bounded frame checks and separate compiler obligations.
Closing CD3, CD4, CD5, CD6	Four distinct tables; constrained symbolic and startup checks.
Shorter-path alternative	Uniform symbolic and startup checks; maximum source path 36.
Rotated196	Uniform symbolic and startup checks.
Rotated macro62	Uniform symbolic and startup checks.
Eight-zero scanner regrouping	A smaller grammar whose scanner repair restores the physical count to 196.
Rebracketed196	Same-size physical source rearrangement.
H-only Shift	Source reduced to 3,811 sites; physical count remains 196.
Later twelve-rotation layout	Additional distinct graph absent from the older fourteen-table census; freshly replayed here.

The later layout is the member `pa_consistency_research_2026-11-13/rotated196/machine.tm` of `pa_consistency_research_2026-11-13(2).zip`. Its table digest is

`b9873b89df4189414d0f443922ca9227bf2f76d8c81a0d0951c06cd608d6a90b`.

Fresh replay reconstructs its twelve literal rotations, checks equality of all 3,812 ordinal source instructions and successors, verifies 4,959 symbolic primitive contracts, and checks the exact blank startup at 58,294 transitions. The table has 196 action-bisimulation classes. These are executable checks relative to the inherited engine contracts, not a fresh full Lean build or an internal PA equivalence theorem. The symbolic evaluator's own step counts must not be confused with literal TM running time.

3 Which reductions combine, and which do not

3.1 The U208 fold is already a different local situation

In the native U209 parent used by the original U208 submissions, state 211 is entered by

$$163, 0 \mapsto (0, L, 211), \qquad 163, 1 \mapsto (1, R, 211).$$

Its one-row behavior can be shared with state 172 using an entry-history invariant and a helper. The compared PA196 table has no state whose zero and one rows have this exact mixed-direction, same-target motif. Therefore a renaming cannot transplant that particular fold. This is a narrow structural obstruction, not a proof that no history-sensitive fold exists in PA196.

3.2 Source size and physical size diverge

Source transformation	Source result	Physical result or limitation
Direct negation peephole	3,596 sites	197 states
Negation plus MP success	3,573 sites	197 states
Factoring the combined source	157 internal nodes	No smaller table; inherited-role estimate 200
Factoring original PA196	157 internal nodes	Loses a needed helper shape; no smaller table
Maximum zero-run five	159 nodes, 3,812 sites	No validated allocation or blank initializer
Residual MP	3,799 sites, 164 nodes	No smaller physical compilation
Diagonal equality vacuity	3,700 sites	199, subsequently rearranged to 198
Distributed equivalent gate	3,773 sites	198 states
Integrated equality witness	3,745 sites	203 states; an earlier layout uses 206
H-only Shift	3,811 sites	196 states
Scanner-aware regrouping	158 internal nodes	A rejected 195-state table repaired to 196

The figures above are the submitted constructions and their stated checks, as indexed in `census/archive_comparison.json`. The current investigation does not rerun every large original symbolic verification.

Three separate costs are involved. The expanded instruction word counts primitive sites. The source grammar counts distinct internal binary nodes. The literal machine additionally pays for dispatch, leaf routing, register arithmetic, source-path scanning, and startup. Shared readers depend on the preceding path bits and on other readers serving as helpers. A source rotation can preserve every ordinal successor while breaking those physical assumptions. Fewer grammar nodes can consequently require more states.

3.3 Every supplied 195-state diagnostic is identified

The five 195-state files are not five independent successes. They represent four candidates:

1. The two-rotation quotient appears in native and dense numbering. Its submitted replay records 93 failed primitive contracts.
2. `warm_physical_61_8047` has 714 recorded symbolic failures. A valid source frame at site 1111 enters an exact four-transition cycle after 26 transitions.
3. `extended_physical_61_12085` has 777 failed contracts in its accompanying candidate receipt. This investigation recounts and identifies the file but does not freshly replay a concrete rejection witness for it.
4. The seven-zero scanner regrouping reaches a wrong clean frame from site 1003 at 276 transitions. With register 1 initially one, the reported 278-transition endpoint also has the wrong logical decrement/branch. Extending the scanner restores the working-state count to 196.

These observations refute the claimed source simulations. They do not by themselves decide each candidate's blank-run halting set or prove that an entirely different consistency interpretation is impossible. The tables and their scoped failure records are retained under `census/rejected195/`.

4 A checked family of separation-gate spellings

The original source constructs the formula

$$S = \forall x_b\big((X \to (P \to Y)) \wedge (Y \to P)\big),$$

where X abbreviates membership $M(a, b)$, Y abbreviates $M(c, b)$, and P denotes the property formula with numeric code c. The code c also serves as a variable label elsewhere in the gate. We do not change that coding convention or the guard $c \neq a, b$.

Define two alternative antecedent arrangements and two conjunction orders:

$$A_0 = X \to (P \to Y), \qquad A_1 = P \to (X \to Y), \qquad B = Y \to P,$$

$$S_{e,0} = \forall x_b(A_e \wedge B), \qquad S_{e,1} = \forall x_b(B \wedge A_e), \qquad e \in \{0, 1\}.$$

Keep the original point formula, equality vacuity Δ, membership vacuity V, and the corresponding quantified clause $Q_{e,j}$ constructed from $S_{e,j}$. There are two outer orders:

$$F_{e,j,0} = \neg(\Delta \to (V \to Q_{e,j})), \qquad F_{e,j,1} = \neg(V \to (\Delta \to Q_{e,j})).$$

This gives eight spellings. The precise numeric constructor and macro words are in `gates/gate_variants.py` and `gates/gate_variants.json`.

Proposition 4.1 (Local equivalence)**.** *All eight displayed gates are classically equivalent with the same decoded parameters. Each separation matrix is equivalent to the original one under every valuation of X, P, Y.*

Proof. The two implications $X \to (P \to Y)$ and $P \to (X \to Y)$ both say $\neg X \vee \neg P \vee Y$. Exchanging conjunction operands preserves the matrix. Replacement of equivalent formulas under a fixed universal quantifier preserves equivalence. Finally, each outer gate says $\Delta \wedge V \wedge \neg Q_{e,j}$. All parameter labels, quantifiers, and freshness guards remain the same. □

The argument is an ordinary finite logical equivalence, not merely an observed numerical match. It can be compiled into fixed propositional derivations and the existing quantifier congruence rules. The packet does not export that compiler as a new Lean or PA proof. The original remaining axiom heads are retained; in particular this argument must not silently assume explosion after deleting the head used to derive it.

The script checks the 32 matrix valuations and the eight outer-order valuations, for 40 finite checks. It emits all eight gate words for both of the inherited global equality-argument conventions. Each word has 77 macro tokens, the same as the original gate. Thus any state saving here would come from improved sharing and routing, not a shorter token word.

4.1 An exact local formula-size calculation

Let the available atoms be $X, P, Y, \bot, \top$ and let implication be the only binary connective. Count occurrences of implication in a formula tree; do not count shared subexpressions as a single occurrence.

Theorem 4.2 (Restricted matrix minimum)**.** *The Boolean function*

$$((X \to (P \to Y)) \wedge (Y \to P))$$

requires at least six implication occurrences in this syntax. Exactly four minimum-size trees express it:

$$((A_e \to (B \to \bot)) \to \bot), \qquad ((B \to (A_e \to \bot)) \to \bot), \qquad e = 0, 1.$$

Proof. For the eight valuations ordered lexicographically by (X, P, Y), represent a truth function by an eight-bit integer. Implication is the operation $(a, b) \mapsto ((\sim a) \mid b)\ \&\ 255$. The target function has mask 157. The size-zero functions are the three variables and the two constants. At size k, enumerate pairs of previously known minimum-size functions whose sizes sum to $k - 1$, form their implication, and retain all minimum-size syntax trees for newly obtained functions.

This recurrence is complete for minimum tree size: if a minimum tree had a nonminimum child for its Boolean function, replacing that child would produce a smaller tree for the same parent function. Therefore every minimum tree appears in the recurrence. The numbers of functions first appearing at sizes 1 through 6 are respectively

$$9,\ 21,\ 31,\ 38,\ 46,\ 55.$$

The explicit finite calculation, reproduced by the included script, first reaches mask 157 at size six, with exactly the four trees displayed above. The two conjunction encodings furnish the upper bound directly. □

This is a small, reproducible formula-size theorem. It is not a theorem that six implications minimize machine states, not a grammar-size lower bound, and not an obstruction to a larger formula whose repeated pieces compile more economically.

4.2 Search with a physical sharing score

The new search varies the eight gate spellings together with the inherited head order, independent schematic label renamings, global equality-argument orientation, and shared binary-tree reassociations. It evaluates a physical reader-sharing score instead of minimizing expanded instructions alone.

The fixed part contains 134 states in the inherited macro allocation. If a candidate AX grammar has n internal readers and its compatible left-entry/right-entry graph has a matching of size f, the discovery score is

$$134 + n - f.$$

The baseline gives $134 + 68 - 6 = 196$. Matching is only a syntactic discovery criterion: simultaneous folds can interfere through their helpers, source paths may violate scanner assumptions, and blank startup must still be checked. The score therefore is neither a certificate nor a lower bound.

The recorded run uses 15,000,000 iteration slots with seed 196195 and records 13,260,073 score evaluations, including the audited seed and restart bookkeeping. Replaying those corrections leaves all eight final grammar witnesses byte-identical. Its best retained, filter-eligible score remains 196. Independent checking of the eight final witnesses gives eligible reader scores 62, 67, 66 and 66 for gates 0, 2, 4 and 6, respectively, hence nominal totals 196, 201, 200 and 200. The other four saved witnesses contain seven or eight consecutive zeros and are penalty diagnostics, not physical estimates. The discovery filter is only a preliminary path constraint; it does not certify runtime.

The independent checker verifies the exact 201-token word, all expanded source paths, reader counts, two independently enumerated helper-edge sets, and maximum matching. It also compares the actual C++ header arrays with the JSON inputs. The baseline expands to 3,812 sites and 4,959 potential primitive branch cases; these branch cases are generated, not newly executed by this grammar-score audit.

The complete source, pinned inputs, final grammar witnesses, random seed and receipts are in `gate_search/`. The separate verifier recomputes the source word and reader cost independently in Python. No candidate with a verified physical count below 196 results from this experiment.

5 Additional physical reduction searches

The existing repository already contains substantial direct state-fold searches. The additional screens here extend specific limits and retain their exact scope. A failed bounded search is not a lower bound on all 195-state machines.

5.1 Completed wider and displaced screens

The radius-six blank-start abstract closure has 270,067 state/window pairs. Its seven-cell projection has 8,120 contexts. It contains an abstract route to halt, so it is used as an overapproximation for local simulation checks, not as a nonhalting certificate.

The wider direct-fold screen checks all 38,220 ordered deleted/retained state pairs. Child segments may last up to 32 transitions and must stay inside seven cells. After projecting the deleted state to the keeper, up to two retained rows may change arbitrarily. It examines 98,072,128 first-row assignments and 798,376,992 second-row assignments. There are no unpatched, one-row or two-row survivors.

The displaced screen uses the full thirteen-cell contexts and offsets -2 or $+2$. Its 33,735 eligible choices are 173 eligible deleted states times 195 keepers. With one additional arbitrary helper-row replacement, 32-step positive segments and exact whole-window endpoints, it examines 47,684 helper locations and 37,384,256 replacement triples. There are no unpatched or patched survivors.

An independent Python simulator checks 512 seven-cell contexts for a known two-helper synthetic fold and 160 five-cell contexts for a genuine displaced fold. The C++ engines also recover these controls. This checks that the new templates can recognize positive examples; it is not a proof that every possible optimization fits either template.

5.2 Three-row screen: a recorded partial result

A recursive failure-trace search handles all choices of up to three changed rows for each selected pair, within five cells and sixteen child steps. This differs from the older representative-only third-row extension. The planned pair set contains the 348 ordered pairs sharing an original outgoing row.

Exactly 84 completed pairs are recorded here. They examine 162,081 first-row, 328,246,128 second-row and 9,582,125,364 third-row assignments, with zero survivors. The run was stopped to pursue source specializations; it is not exhaustive for the 348-pair set, much less for all state pairs. The exact completed pairs are in `physical_search/partial_three_receipt.json`.

5.3 A shorter H-only Shift guard

The accepted H-only Shift source enters its comparison with scratch register $X = 1$ and arbitrary H. Under that explicit condition, the generic eleven-site equality guard can be specialized to seven sites: clear X; test H twice; halt precisely if its entry value was one; otherwise clear the remaining H and continue. The cases $H = 0$, $H = 1$ and $H \geq 2$, followed by induction on the clearing loop, prove the all-natural local contract.

The independent source checker resolves both full control graphs twice, checks the exact seven-row replacement, verifies all 3,800 exterior rows under the ordinal shift, and rules out exterior entries into the removed interior. The H-only source decreases from 3,811 to 3,807 sites. However, its distinct internal nodes increase from 159 to 163, because the old comparator's pieces are reused elsewhere. The first leaf shortens from `000000` to `00000`, but twenty other addresses still contain six consecutive zeros. Maximum depth remains 39. No shortened scanner, binary allocation or new blank-start proof follows. The exact source and ordinary proof are included in `physical_search/`.

5.4 Why displaced projection is a legitimate search family

Ordinary state identification maps (p, h, T) to (q, h, T). A displaced fold instead maps

$$(p, h, T) \longmapsto (q, h - 2, T)$$

when every external incoming transition to p moves right. If a predecessor would move from $h - 1$ to h, reversing that incoming move to left brings the child to $h - 2$, exactly the projected position. The reflected case uses a $+2$ offset and reverses uniformly left-moving entries.

Lemma 5.1 (Conditional positive-block transfer)**.** *Suppose a configuration relation maps the initial configurations of two deterministic machines with total working-state transition tables and preserves the halting predicate. Suppose each parent transition from a reachable nonhalt configuration is matched by a positive finite child segment ending at the related next configuration, with no intermediate child halt. Then the two blank runs halt together. If the relation, its invariant and the positive-segment existence facts are proved uniformly in PA for arbitrary internally coded reachable configurations and represented finite run codes, the halting equivalence is provable in PA.*

Proof. Concatenate the matching positive segments. A parent halt gives a child halt at the corresponding endpoint. If the parent never halts, positivity makes the cumulative child lengths unbounded; every child time belongs to a nonhalting segment or endpoint, so the child never halts. The finite induction on coded runs gives the internal version directly: if the child halts at time t and the parent has not halted in its first t steps, lifting those steps yields a nonhalting child prefix of length at least t, contradicting trace determinism. The case $t = 0$ follows from initial halting preservation. The lemma does not supply those representation and invariant proofs for a new concrete table. □

This explains the purpose of exact whole-tape endpoint checks and positive segment lengths in the displaced screen. It also explains why agreeing on a long blank-tape prefix would be inadequate. Even a successful search receipt would still need a verified invariant covering every relevant configuration and a binding to the actual parent machine.

6 The arithmetic proof boundary

The precise imported theorem inventory [2] distinguishes:

$$\text{a Lean-metatheoretic theorem of literal nonhalting,} \qquad \mathrm{PA} \vdash \mathrm{Nonhalt}(M_{196}) \to \mathrm{Con}(\mathrm{PA}).$$

Its standard-model and unprovability declarations do not change the latter formula into a biconditional. This report does not claim that the converse is false or unprovable. It identifies the statements actually exported and the proof obligations explicitly recorded in the repository comparison and reverse-bridge notes [3, 4].

The missing direction needed for the requested endpoint can be written as

$$\mathrm{PA} \vdash \mathrm{Halt}(M_{196}) \to \neg\, \mathrm{Con}(\mathrm{PA}).$$

A concrete uniform route has two main stages.

First, from an arbitrary internally coded finite literal halting trace, extract a refutation in the exact numeric source calculus. This requires the startup binding, positive nonhalt macro segments, internal trace determinism, source scheduler extraction, and the proof-bearing store and stack invariant. Ordinary standard-natural simulations do not alone prove the arbitrary internal-time statement. In particular, the numeric stack uses $\mathrm{pair}(0,0) = 0$, so an empty stack and a pushed zero with empty tail have the same numeric code. A proof-carrying invariant must explicitly accommodate the zero alias rather than assume injectivity of the list representation.

Second, translate the resulting native proof code into a PA refutation. The direct proposed route interprets membership by arithmetic bit membership, uses arithmetic equality, translates the transposed quantifier convention, and builds bounded-comprehension instances using a fixed induction template. It must handle variable-label collisions, the original code/label freshness convention, the special truth alias at code zero, substitution, and the exact target PA proof coding. Proof-producing syntactic transformations, verified uniformly, avoid asserting a truth predicate for all arithmetic formulas.

These stages are a concrete program of work, and earlier archives supply substantial ordinary mathematical arguments for pieces of it. They are not completed by the new finite comparisons here. Likewise, a successful halting-preserving reduction of PA196 would inherit whatever endpoint has actually been proved for PA196; it would not create a missing endpoint theorem merely by reducing the state count.

7 What transfers from ZFC and concise iPRA

The useful ZFC contributions are methods [6]: opposite comparison residuals, history-sensitive dispatch, reallocation of source-tree readers, and symbolic primitive contracts over arbitrary register values. Applying a source identity alone can increase the grammar cost. The incoming residual-MP construction is an example: it reduces PA's source to 3,799 primitive sites while raising the internal-node count to 164.

The concise-iPRA repository [7] concerns a different checker and a different formal theory. A shorter checker for iPRA does not automatically provide a search for PA refutations or a PA-provable equivalence with Con(PA). To use its coding ideas here, one would need the uniform PA proof translations and an actual binary compilation with all constants, startup and dispatch charged to the state count. The imported PA/iPRA interface already exposes endpoint assumptions; it does not eliminate them.

An additional explored idea reverses the implication constructor's pair arguments while retaining truth code zero and contradiction code one. The direct field-reading macro decreases from 67 to 37 primitive sites. Its contract, for zero scratch registers, is

$$(T, D, L, S, R) = (t, d, \mathrm{pair}(x, \ell), 0, 0) \longmapsto (4\,\mathrm{pair}(d + x, t), d + x, \ell, 0, 0).$$

Replacing the two old occurrences removes 60 sites, but increases the source DAG from 159 to 162 nodes. The AX prefix uses both the old parameter and its subsequently incremented value. The included corrected prefix preserves that schedule and uses an additional reversed stack-pair macro. Its inherited association is costly and no smaller physical compilation results. The ordinary loop proofs and exact subtrees are under `theory/`.

7.1 Membership orientation and its precise proof limitation

Membership transposition is a cleaner emitter experiment:

$$M_0(a, b) = 4\,\mathrm{pair}(a, b) + 1, \qquad M_1(a, b) = 4\,\mathrm{pair}(b, a) + 1.$$

Implication and the arithmetic MP routine remain unchanged, as do truth code zero, the false target $1 = M_0(0, 0) = M_1(0, 0)$, and the 18-token AX prefix. Combining this choice with equality orientation and the eight gate spellings gives 32 mode/gate combinations. The independent audit decodes emitted words into free constructor trees and verifies all 248 head/gate orientation identities, in addition to the generator's 1,792 numeric checks.

For each orientation, define $R(0) = 0$, recurse on both implication children and the quantified body, keep binder labels fixed, and swap the two literal labels of an atom exactly when its orientation changes. Structural induction proves

$$\mathrm{decode}_{\mathrm{new}}(R(n)) = \mathrm{decode}_{\mathrm{old}}(n), \qquad R(R(n)) = n.$$

Names and free-name sets are preserved. Every name in a decoded positive code is strictly below that code, by the Cantor-component bounds. Both old and new private code labels are consequently fresh for their common decoded property.

These lemmas do *not* make the gate schema commute literally with recoding. In the membership-transposed convention, $R(5) = 9$. The original gate parameters $c = 5, a = 9, b = 0$ satisfy $c \neq a, b$, but replacing the property code by 9 makes the emitter's new guard fail. Recoding the old gate retains binder 5, whereas the new emitter uses binder 9. An actual alpha-renaming or padded-property proof compiler is therefore required. The packet supplies the recoding lemma and this counterexample, not that missing compiler.

The four-mode search uses 16,000,000 iteration slots with seed 3196195 and records 14,142,234 score evaluations. The audited replay leaves all 32 final grammar witnesses byte-identical. The best retained score remains the original 196. Among the saved filter-eligible witnesses, the best scores in modes 0, 1, 2 and 3 are respectively 196, 200, 202 and 205; modes 2 and 3 are the membership-transposed conventions. These are nominal allocation scores, not verified machines. All 32 final witnesses are checked independently against their exact emitted constructor trees and helper matching. Thus this additional coding family produces neither a smaller physical table nor a completed logical transport.

7.2 A rejected logical shortcut

Another tempting change replaces the first head $(\exists d\, \forall d\, H) \to H$ by $\neg H \to \forall d\, \neg \forall d\, H$. Classical contraposition is insufficient justification here: native $\neg A$ is $A \to 1$, and the original first head is used to derive explosion for code 1.

There is a concrete counterinterpretation to derivational equivalence. Interpret implication as Boolean implication, every universal quantifier as the identity, every equality atom as false, and $M(a, b)$ as true exactly when its two literal labels coincide. Both code zero and code one are then true. All relative negations, gates, equality-existence heads and the proposed replacement head are true. The membership-transport heads have a false equality antecedent; the other quantifier heads and propositional basis are also valid. MP and generalization preserve truth. But $H = M(0, 1)$ is false, and the old first head with this H is false. Induction on derivations rules out deriving that old head in the replacement calculus. This refutes the suggested equivalence proof; it does not independently determine the consistency strength of every alternative theory.

The full all-parameter argument and a finite implementation check are in `theory/THEORY.md` and `theory/head1_countermodel.py`.

8 Verification and reproducibility

The portable checks use explicit file identities and declared trust scopes. The quick entry point is

```
python3 verify_all.py
```

It verifies the final byte manifest, the fifteen-table census and U208 comparison, the eight gate spellings and restricted formula minimum, and the independent grammar-score witnesses. It does not silently rerun every long search, rebuild all imported Lean projects, or claim a PA consistency theorem. The longer physical screens have their own commands and receipts.

To rebuild the PDF, run twice from this directory:

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The designated 196-state, U208 and rejected 195-state transition-table corpora are included; the remaining archival inputs are pinned by filenames, member paths and digests in the census. Original authorship and licenses are retained. The imported projects were read as pinned reference trees; the original repository's machine tables and theorem statements were not changed, and no branch, commit, PR or message was published.

9 Further research questions

1. Can a joint physical allocator exploit a five-zero source representation while preserving the return scanner, entry-history aliases and a new blank initializer? Optimizing only the source grammar has repeatedly missed these costs.

2. Can a slightly larger equivalent separation formula provide more useful shared constructor fragments? The six-implication minimum proved here does not answer this physical question.

3. Can a reversed implication constructor use a saved-old-parameter choreography with fewer net reader and engine states? The AX prefix and MP comparison must be redesigned together.

4. Can displaced state identification use a stronger source-conditioned invariant, a larger offset, or several coordinated states? The present finite-window screens cover only their stated templates.

5. Can the arbitrary-internal-time literal-to-source extraction and the uniform native-to-PA proof compiler be completed independently of the state search? This would make the exact consistency biconditional reusable for every later certified physical reduction.

A All top-level incoming archives

The names below are exact repository filenames. Full digests, sizes and member counts are in `census/archive_bytes_inventory.json`; their contributions and proof-status qualifications are in `census/archive_comparison.json`.

Archive	Contribution and limitation
`Below_U208_Research_2026-10-29.zip`	Comparison and scoped local/source/offset screens; no machine below 208 states.
`ConPA_below208_research_2026-10-29.zip`	Source permutations, two-row screens, logical obstruction; no machine below 208 states.
`PA_consistency_TM_research_2026-11-13.zip`	Fourteen-table census; original 3,812-site source: zero-run-five, 159-node grammar; combined 3,573-site source: 157-node grammar; residual MP 3,799-site source, 164 nodes; no new physical 196 or 195 table.
`U208_ConPA_2026-10-29.zip`	Original U208, class A; same 172/211 fold as companion submissions.
`U208_ConPA_Conditional_Context_2026-10-29.zip`	Original U208, classes A/B; conditional-history proof presentation.
`U208_ConPA_Entry_Context_2026-10-29.zip`	Original U208, classes A/B; entry-context proof presentation.
`U208_ConPA_History_2026-10-29.zip`	Original U208, classes A/B; predecessor-history proof presentation.
`conpa_tm_comparison_and_reduction_search.zip`	Two experimental 196 MP variants; bounded frame checks; rejected Shift halts after 43,563 steps.
`conpa_tm_comparison_and_search.zip`	A/D swap 196; rotated 195 rejected; unmodified scanner shortcut halts after 699 steps.
`pa196_research_2026-11-13.zip`	157-node source, no binary table; missing-helper obstruction; broader two-row synthesis negative.
`pa208_research_2026-10-29.zip`	Eighteen further pairwise nonisomorphic 208 helper alternatives.
`pa_consistency_comparison_2026_11_13.zip`	Quantifier-head splitting with Hilbert certificates; 12 combined source/table experiments; best combined negation/MP 197; newer gate and padding searches; no smaller machine.
`pa_consistency_reduction.zip`	Rebracketed 196, H-only Shift 196, integrated witness compiler 203/206.
`pa_consistency_research.zip`	158-node source compiled to 196 states after eight-zero scanner repair; no machine below 196 states.
`pa_consistency_research_2026-10-29.zip`	Source: 3,909 to 3,904 sites; DAG: 197 to 200 nodes; no smaller table.
`pa_consistency_research_2026-11-13(2).zip`	Additional distinct 196-state twelve-rotation layout; diagonal-vacuity 198; halt-at-successful-MP command theorem; reduced joint register abstractions.
`pa_consistency_research_2026-11-13.zip`	Diagonal-vacuity 199; negation peephole 197; no machine below 196 states.
`pa_tm_comparison_and_reductions.zip`	Two rotated 196; equivalent gate compiler 198 with 3,773 sites.
`pa_tm_extended_research_2026-11-13.zip`	Four 196 closing-CD variants with constrained symbolic/startup checks.
`pa_tm_research_2026-11-13.zip`	Alternative 196 with maximum source path 36 and full symbolic/startup checks.
`pa_unary_power_cover.zip`	Unary-cover theory and proof stream; no binary compilation.
`u208_comparison_research_2026-10-29.zip`	Rejected 207 with concrete primitive-frame counterexample.
`u208_research_2026-10-29.zip`	Original U208, class B primary and class A alternative.

References

[1] James Chang, *Nonhalting and PA-independence for a 196-state Turing machine*, imported PA196 revision `407cdc356f6b586635e71ac92be394255f4bfea4`, `https://github.com/jamesyc/BusyBeaver/blob/407cdc356f6b586635e71ac92be394255f4bfea4/paper/pa196.tex`. The exact consulted copy is ProveIt's `Computability/BusyBeaver/Research/PA/jamesyc-upstream/pa196/paper/pa196.tex`.

[2] James Chang, PA196 formal theorem statements, `docs/result.md` in the same imported snapshot; retained in this packet as `reference/pa196_theorem_inventory.md`.

[3] ProveIt, *PA196 and the incoming U208 construction*, `Computability/BusyBeaver/Research/PA/pa196-vs-u208/COMPARISON.md`, commit `3268f238cb6be4b9c68b6cddaf289f173f77566e`; consulted content retained in `reference/`.

[4] ProveIt, *PA196: a route to the missing internal implication*, `Computability/BusyBeaver/Research/PA/pa196-vs-u208/PA-BRIDGE.md`, same commit; consulted content retained in `reference/`.

[5] ProveIt, the 23 incoming archives listed in Appendix A, with exact member provenance and digests in the supplied census. Repository: `https://github.com/VladimirReshetnikov/ProveIt/tree/3268f238cb6be4b9c68b6cddaf289f173f77566e/docs/incoming`.

[6] ProveIt, ZFC consistency-machine research, `Computability/BusyBeaver/Research/ZFC`, at the same commit.

[7] ProveIt, concise intuitionistic primitive recursive arithmetic, `Logic/PrimitiveRecursiveArithmetic/ConciseIPRA`, at the same commit.